

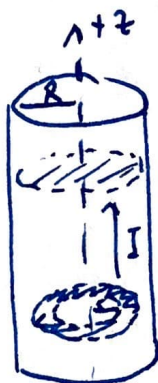
Question 1

A long, straight, solid cylindrical conductor of radius R carries a steady total current I in the $+\hat{z}$ direction. The volume current density varies with radial distance s from the axis as

$$\vec{J}(s) = \begin{cases} J_0 \frac{s}{R} \hat{z}, & 0 \leq s \leq R, \\ 0, & s > R. \end{cases}$$

- (a) Determine J_0 in terms of the total current I and the radius R . — 0.4 pts
- (b) Using Ampère's law, find the magnetic field $\vec{B}(s)$ (magnitude and direction) for
- (i) $s < R$. — 0.2 pts
- (ii) $s > R$. — 0.2 pts
- (c) Sketch $B(s)$ versus s for $0 \leq s \leq 2R$. — 0.2 pts

Solution:



a)

$$I = \int \vec{J}(s) \cdot d\vec{a} = \int_0^R J_0 \frac{s}{R} 2\pi s ds = \frac{J_0}{R} 2\pi \int_0^R s^2 ds$$

$$= \frac{J_0}{R} 2\pi \frac{R^3}{3} = \frac{2\pi J_0 R^2}{3} \Rightarrow J_0 = \frac{3I}{2\pi R^2}$$

b) $s < R$:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc} \rightarrow B 2\pi s = \mu_0 \left(\int_0^s J_0 \frac{s'}{R} 2\pi s' ds' \right)$$

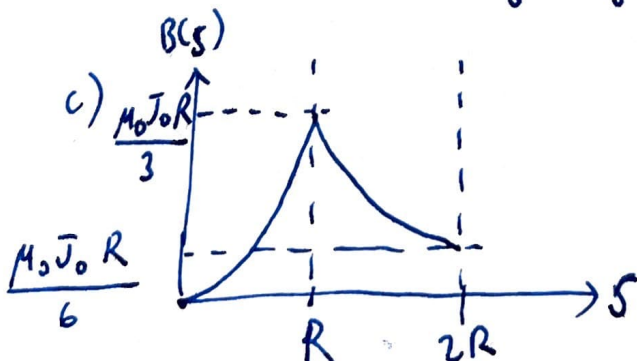
$$B 2\pi s = \mu_0 \frac{J_0}{R} 2\pi \frac{s^3}{3} \Rightarrow B = \frac{\mu_0 J_0}{3R} s^2$$

in $\hat{\phi}$ direction.

$s > R$:

$$B 2\pi s = \mu_0 \frac{2\pi J_0 R^2}{3}$$

$$B = \frac{\mu_0 J_0 R^2}{3s} \text{ in } \hat{\phi} \text{ direction}$$



Question 2

A long solenoid, of radius a , is driven by an alternating current, so that the magnetic field inside the solenoid is sinusoidal:

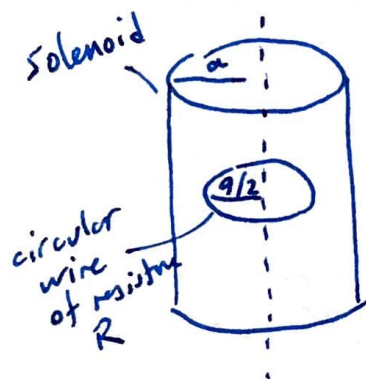
$$\vec{B}(t) = B_0 \sin(\omega t) \hat{z}$$

where B_0 is the amplitude of the magnetic field, ω is the angular frequency, and $\hat{z}$ is the unit vector in the direction of the solenoid's axis. A circular loop of radius $a/2$ and resistance R , is placed inside the solenoid, and is coaxial with it.

(a) Find the current induced in the loop as a function of time. — 0.5 pts

(b) Find the power dissipated in the loop as a function of time. — 0.5 pts

Solution:



$$a) \quad \mathcal{E} = - \frac{d\Phi}{dt} \quad \Phi(t) = B_0 \sin(\omega t) \pi \frac{a^2}{4}$$

$$= -B_0 \omega \cos(\omega t) \pi \frac{a^2}{4}$$

$$\mathcal{E} = IR \rightarrow I(t) = \mathcal{E}/R = - \frac{B_0 \omega \pi a^2}{4R} \cos(\omega t)$$

$$b) \quad P(t) = [I(t)]^2 R = \frac{B_0^2 \omega^2 \pi^2 a^4}{16 R^2} \cos^2(\omega t) R$$

a) As $\vec{B}(t)$ increases in $+\hat{z}$ direction the current is clockwise
 $\vec{B}(t)$ decreases in $+\hat{z}$ " " " counter clockwise
 $\vec{B}(t)$ inc. in $-\hat{z}$ " " " " counter clockwise
 $\vec{B}(t)$ dec. in $-\hat{z}$ " " " " clockwise

Question 3

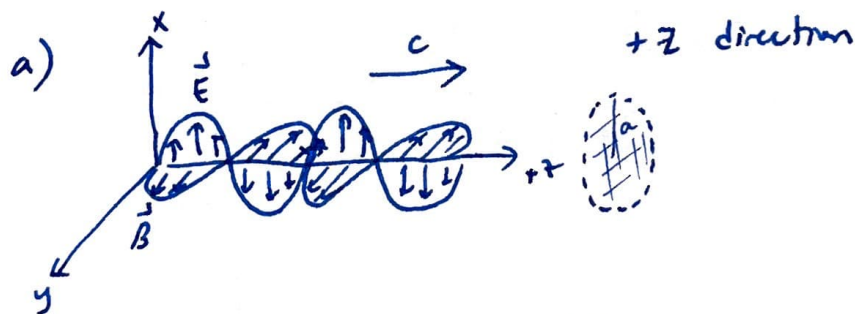
The magnetic field component of an monochromatic electromagnetic wave propagating in free space is given by plane waves as

$$\vec{B}(z, t) = B_0 \sin(kz - \omega t) \hat{y}.$$

- What is the direction of propagation of the wave? — 0.1 pts
- Find the electric field component of the wave. — 0.2 pts
- Find the Poynting vector of the wave. — 0.2 pts
- How much power is carried by the wave through a circular area of radius a in the plane perpendicular to the direction of propagation? — 0.2 pts
- What is the direction of polarization of the wave? — 0.2 pts
- If the wave goes through a polarizer oriented at an angle $\pi/4$ with respect to the direction of polarization, what would be the amplitude of the transmitted wave? — 0.1 pts

Given: $\vec{B}_0 = \frac{1}{c}(\hat{z} \times \vec{E}_0)$ where c is the speed of light in vacuum.

Solution:



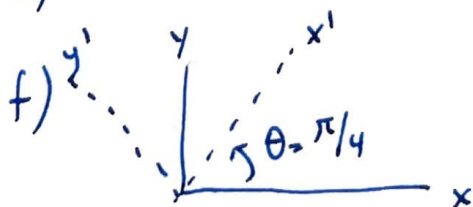
b) $\vec{E}(z, t) = c B_0 \sin(kz - \omega t) \hat{x}$

c) $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} = \frac{c B_0^2}{\mu_0} \sin^2(kz - \omega t) \hat{z}$

d) Def. of $\vec{S}$: Energy carried by em wave per unit time per unit area.
(or power carried by em wave per unit area)

e) $P(z, t) = \frac{c B_0^2}{\mu_0} \sin^2(kz - \omega t) \pi a^2$

f) $\hat{x}'$ direction



$\vec{E}(z, t) = c B_0 \sin(kz - \omega t) \frac{1}{\sqrt{2}} (\hat{x}' - \hat{y}')$

=> After the polarizer:

$\vec{E}'(z, t) = \left(\frac{c B_0}{\sqrt{2}} \right) \sin(kz - \omega t) \hat{x}'$
→ amplitude.

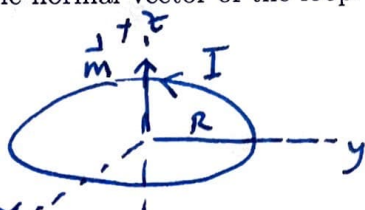
this part will be blocked

Question 4

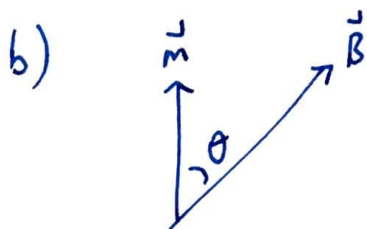
A circular loop of radius R carries a steady current I .

- (a) Write down the magnetic dipole moment $\vec{m}$ of the loop in terms of I and R . — 0.4 pts
- (b) Now, place the loop in a uniform external magnetic field $\vec{B}$ that points at an angle θ with respect to the normal vector of the loop. Write the torque $\vec{\tau}$ acting on the loop. — 0.6 pts

Solution:



a) $\vec{m} = I \pi R^2 \hat{z}$



$\vec{\tau} = \vec{m} \times \vec{B}$: The direction of the torque will be somewhere in the xy plane according to the choice of $\vec{B}$.

$$|\vec{\tau}| = |\vec{m}| |\vec{B}| \sin \theta$$

$$= I \pi R^2 B \sin \theta$$